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Assignment -4
Artificial Intelligence

Q1. State Space Representation (Case Study) A logistics company is trying to optimize its delivery system in a smart city. The city is represented as a graph where nodes represent locations and edges represent routes with associated costs (distance, traffic, fuel consumption). The system must determine the most efficient route for delivery vehicles while considering constraints like traffic congestion and delivery deadlines. Tasks:

1.) Model the problem using state space representation (define state, operators, initial state, goal state).

2.) Explain how search algorithms like

Breadth-First Search, Depth-First Search, and A* can be applied.

3.) Compare uninformed vs informed search strategies in this context.

4.) Suggest the most suitable approach and justify your answer.

Ans.) 1. State Space Model
State: A state represents the current location of the delivery vehicle + remaining deliveries + current time/traffic condition. Example: (current_location, pending_deliveries, time)

Operators (Actions): Move from one node (location) to another via available routes (edges).

Initial State: Vehicle at depot with all deliveries pending.

Goal State: All deliveries completed within deadlines (optimal route found).

Cost Function: Combination of distance, fuel consumption, and traffic delay.

Applying Search Algorithms

Breadth-First Search (BFS): Explores level by level.

Guarantees shortest path (in terms of steps, not cost).

Not suitable here because costs vary (traffic, fuel).

Depth-First Search (DFS): Explores deep paths first.

Low memory usage

May find suboptimal or very poor routes.

A* (Best choice among these):
Uses: actual cost so far : heuristic (e.g., estimated distance to destination)

Efficiently finds optimal routes using heuristics like:

straight-line distance

predicted traffic

3. Uninformed vs Informed Search
Aspect
Uninformed (BFS, DFS)
Informed (A*)
Knowledge
No domain knowledge
Uses heuristics

Efficiency
Low
High
Optimality

Limited
High (if heuristic admissible)

Use Case
Simple graphs
Real-world routing

4. Best Approach

A* is most suitable Why?
Handles weighted costs (distance + traffic)
Scalable for smart cities
Can integrate real-time data (traffic APIs)

Q2. Reinforcement Learning (Case

Study) A self-driving car company is developing an AI agent to learn how to drive safely in urban environments. The car receives rewards for safe driving and penalties for collisions, traffic violations, or inefficient routes. Tasks:

1.) Formulate the problem as a Reinforcement Learning (RL) model (states, actions, rewards, environment).

2.) Explain how Q-learning or Deep Reinforcement Learning can be used.

3.) Discuss the exploration vs exploitation trade-off in this scenario.

4.) Identify challenges such as sparse rewards, safety concerns, and real-world deployment.

Ans.)

1. RL Model

States: Car position, speed, nearby vehicles, traffic signals

Actions: Accelerate, brake, turn, change lane

Rewards:
Safe driving
Reaching destination quickly –
Collisions – Traffic violations – Fuel inefficiency
Environment: Urban road system with dynamic traffic

2. Q-Learning / Deep RL
Q-Learning: Updates value:
Deep RL (DQN): Uses neural networks to approximate Q-values Useful for:
large state spaces vision-based driving (camera input)

3. Exploration vs Exploitation
Exploration: Try new actions (riskier

but learning)
Exploitation: Use known best actions (safer)

In self-driving:
Too much exploration = dangerous
Solution: safe exploration (simulation training first)

4. Challenges
Sparse rewards (few success signals)
Safety-critical decisions
Real-world unpredictability
Sim-to-real transfer problem

Q3. Neural Network Learning & Genetic Algorithms (Case Study) A healthcare organization wants to develop a predictive system for early detection of heart disease. They are considering two approaches: Neural Networks and Genetic Algorithms.

Tasks:

1.) Explain how a Neural Network model can be designed for prediction (architecture, training, backpropagation).

2.) Describe how Genetic Algorithms (GA) can optimize feature selection or model parameters.

3.) Compare both approaches in terms of accuracy, interpretability, and computational cost.

4.) Suggest a hybrid approach combining Neural Networks and Genetic Algorithms.

Ans.) 1. Neural Network Design
Input Layer: Patient data (age, BP, cholesterol)
Hidden Layers: Feature extraction

Output Layer: Probability of heart disease

Training:
Loss function (e.g., cross-entropy)
Backpropagation updates weights

2. Genetic Algorithms (GA)
Used for: Feature selection (best medical indicators)
Hyperparameter tuning

Steps:
Initialize population (feature subsets)
Evaluate fitness (model accuracy)
Selection
Crossover
Mutation
Repeat

3. Comparison
Aspect
Neural Networks

Genetic Algorithms
Accuracy
High
Moderate
Interpretability
Low
Medium
Speed
Fast after training
Slower

Use
Prediction
Optimization

4. Hybrid Approach

Combine both:
GA selects best features
Neural Network performs prediction

Benefits:
Improved accuracy

Reduced overfitting
Efficient model

Q4. Natural Language Processing (NLP) (Case Study) A multinational company wants to build a chatbot that can handle customer queries in multiple languages and understand user sentiment.

Tasks:

1.) Explain how NLP techniques like tokenization, stemming, and parsing can be used.

2.) Describe the role of language models and transformers in chatbot development.

3.) Discuss challenges like ambiguity, sarcasm, and multilingual processing.

4.) Propose a system architecture for the chatbot.

Ans.) 1. NLP Techniques
Tokenization: Split text into words
Stemming: Reduce words (e.g., "running" → "run")
Parsing: Understand grammar structure

2. Language Models & Transformers
Modern chatbots use transformers
Examples:
BERT (understanding)
GPT (generation)
Advantages:
Context awareness
Multilingual capability
Better conversation flow

3. Challenges
Ambiguity ("bank" = river or finance)
Sarcasm detection
Multiple languages

Cultural context differences

4. Proposed Architecture
Input processing (speech/text)
Language detection
NLP preprocessing
Transformer model
Sentiment analysis module
Response generation
Output (text/voice)

Q5. Machine Learning (ML) (Case Study)
An e-commerce company wants to improve its recommendation system to increase sales and customer satisfaction. Tasks: Identify the type of ML approach (supervised, unsupervised, reinforcement learning). Explain how collaborative filtering and content-based filtering can be applied. Discuss data preprocessing, feature engineering, and evaluation metrics.

Ans.) ML Approach
Supervised Learning: Predict ratings
Unsupervised Learning: Customer segmentation
Reinforcement Learning: Adaptive recommendations
Most systems use hybrid approach

2. Recommendation Techniques
Collaborative Filtering:
Based on user behavior
"Users like you also bought..."
Content-Based Filtering:
Based on product features
"Similar items"

3. Data Processing
Preprocessing:
Remove noise
Handle missing values
Feature Engineering:
User preferences
Purchase history